CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

DOUBLE L & N DAIRY

28471 W HULEN RD

GUSTINE CA 95322

NOTES:

PREPARED BY:

CARDOSO AG SERVICES
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Double L & N Dairy

Physical address of dairy:

28471 W Hulen RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1979

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X060-X025-XXXX

B. OPERATORS

Lima, Augusto & Celeste

Operator name: Lima, Augusto & Celeste

Telephone no.: Landline: ; Cellular: (209) 564-1053

28471 W. Hulen RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

Nunes, Alfred or Lucy

Operator name: Nunes, Alfred or Lucy

Telephone no.: Landline: (209) 854-2307; Cellular: (209) 564-1954

28471 W. Hulen RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Lima, Augusto & Celeste

Legal owner name: Lima, Augusto & Celeste

Telephone no.: Landline: ; Cellular: (209) 564-1053

28471 W. Hulen RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

Nunes, Alfred or Lucy

Legal owner name:	Nunes, Alfred or Lucy	Telephone no.:	(209) 854-2307	(209) 564-1954
			Landline	Cellular
28471 W. Hulen RD	Gustine		CA	95322
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	150	205	0	0
Maximum number	0	0	150	205	0	0
Average number	0	0	150	205	0	0
Avg live weight (lbs)	0	0	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 3,608.77 *tons per reporting period*

Total nitrogen from manure: 33,689.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 23,582.65 *lbs per reporting period*

Total phosphorus from manure: 5,701.30 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 9,000,000 *gallons*

Total nitrogen generated: 23,588.83 *lbs*

Total phosphorus generated: 8,517.28 *lbs*

Total potassium generated: 68,521.30 *lbs*

Total salt generated: 267,786.88 *lbs*

	9,000,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	9,000,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Alfreds	40	40	1	none	X070-X070-X019-XXXX
Home	60	60	2	process wastewater	X070-X060-X025-XXXX
Salsa Ranch	78	78	1	manure	X070-X030-X012-XXXX X070-X030-X017-XXXX X070-X030-X018-XXXX
Turkey 3/4	30	30	2	manure	X070-X050-X004-XXXX
Totals for areas that were used for application	168	168	5		
Totals for areas that were not used for application	40	40	1		
Land application area totals	208	208	6		

B. CROPS AND HARVESTS

Alfreds

Field name: Alfreds

11/01/2015: Almond, in shell

Crop: Almond, in shell Acres planted: 40 Plant date: 11/01/2015

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/31/2023	80.00 *ton*	Dry-weight		5.0	22,100.00	1,400.00	22,600.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	2.00	83.98	5.32	85.88	266.00

Home

Field name: Home

Home

11/01/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 60 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	820.00 *ton*	Dry-weight		65.3	20,100.00	3,400.00	28,000.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	13.67	190.64	32.25	265.57	1,251.98

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	1,114.00 *ton*	Dry-weight		54.0	9,300.00	2,500.00	12,300.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	18.57	158.86	42.70	210.10	819.90

Salsa Ranch

Field name: Salsa Ranch

01/15/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 78 Plant date: 01/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	573.00 *ton*	Dry-weight		10.7	34,500.00	2,800.00	23,700.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.35	452.65	36.74	310.95	1,102.10

Turkey 3/4

Field name: Turkey 3/4

11/01/2022: Wheat Hay

Crop: Wheat Hay Acres planted: 30 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	131.00 *ton*	Dry-weight		10.0	20,100.00	3,400.00	28,000.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.37	157.99	26.72	220.08	1,037.52

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	577.00 *ton*	Dry-weight		54.0	9,400.00	2,500.00	12,300.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	19.23	166.33	44.24	217.64	849.34

NUTRIENT BUDGET

A. LAND APPLICATIONS

Alfreds - 11/01/2015: Almond, in shell

Field name: Alfreds

Crop: Almond, in shell

Plant date: 11/01/2015

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.56	0.00	0.00	159.10	3,813,030.00 *gal*
Application event totals		0.56	0.00	0.00	159.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	81.59	1,955,400.00 *gal*
Application event totals		0.29	0.00	0.00	81.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	81.59	1,955,400.00 *gal*
Application event totals		0.29	0.00	0.00	81.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.42	0.00	0.00	119.66	2,867,920.00 *gal*
Application event totals		0.42	0.00	0.00	119.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.56	0.00	0.00	160.46	3,845,620.00 *gal*
Application event totals		0.56	0.00	0.00	160.46	

Alfreds - 11/01/2015: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.44	0.00	0.00	125.10	2,998,280.00 *gal*
Application event totals		0.44	0.00	0.00	125.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.43	0.00	0.00	122.38	2,933,100.00 *gal*
Application event totals		0.43	0.00	0.00	122.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.42	0.00	0.00	121.02	2,900,510.00 *gal*
Application event totals		0.42	0.00	0.00	121.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.38	0.00	0.00	108.79	2,607,200.00 *gal*
Application event totals		0.38	0.00	0.00	108.79	

Home - 11/01/2022: Wheat, silage, soft dough

Field name: Home

Crop: Wheat, silage, soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.79	22.19	226.25	1,063.99	2,250,000.00 *gal*
Canal	Surface water	1.08	0.00	0.00	308.23	11,080,600.00 *gal*
Application event totals		94.87	22.19	226.25	1,372.21	

Home - 11/01/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.57	45.56	372.83	1,730.54	3,150,000.00 *gal*
Canal	Surface water	0.74	0.00	0.00	212.13	7,626,060.00 *gal*
Application event totals		93.32	45.56	372.83	1,942.68	

Home - 05/20/2023: Corn, silage

Field name: Home

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.05	0.00	0.00	300.97	10,819,880.00 *gal*
Application event totals		1.05	0.00	0.00	300.97	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.03	18.23	149.13	692.22	1,260,000.00 *gal*
Canal	Surface water	1.87	0.00	0.00	533.05	19,162,920.00 *gal*
Application event totals		38.89	18.23	149.13	1,225.27	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.61	26.91	189.33	469.41	1,125,000.00 *gal*
Canal	Surface water	0.86	0.00	0.00	246.58	8,864,480.00 *gal*
Application event totals		82.48	26.91	189.33	715.99	

Home - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.14	0.00	0.00	325.45	11,699,810.00 *gal*
Application event totals		1.14	0.00	0.00	325.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.14	29.07	204.47	506.96	1,215,000.00 *gal*
Canal	Surface water	0.99	0.00	0.00	283.75	10,200,670.00 *gal*
Application event totals		89.14	29.07	204.47	790.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.59	0.00	0.00	167.71	6,029,150.00 *gal*
Application event totals		0.59	0.00	0.00	167.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.65	0.00	0.00	185.84	6,680,950.00 *gal*
Application event totals		0.65	0.00	0.00	185.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	270.15	9,711,820.00 *gal*
Application event totals		0.95	0.00	0.00	270.15	

Salsa Ranch - 01/15/2023: Alfalfa, hay

Field name: Salsa Ranch

Crop: Alfalfa, hay

Plant date: 01/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Salsa Ranch - 01/15/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	290.20	94.08	144.44	0.00	1,600.00 *ton*
Application event totals		290.20	94.08	144.44	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	308.23	14,404,780.00 *gal*
Application event totals		1.08	0.00	0.00	308.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	342.40	16,001,690.00 *gal*
Application event totals		1.20	0.00	0.00	342.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.00	0.00	0.00	285.91	13,361,900.00 *gal*
Application event totals		1.00	0.00	0.00	285.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.07	0.00	0.00	306.83	14,339,600.00 *gal*
Application event totals		1.07	0.00	0.00	306.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.40	0.00	0.00	399.58	18,674,070.00 *gal*
Application event totals		1.40	0.00	0.00	399.58	

Salsa Ranch - 01/15/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.82	0.00	0.00	520.22	24,312,140.00 *gal*
Application event totals		1.82	0.00	0.00	520.22	

Turkey 3/4 - 11/01/2022: Wheat Hay

Field name: Turkey 3/4

Crop: Wheat Hay

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	212.21	68.80	105.62	0.00	450.00 *ton*
Application event totals		212.21	68.80	105.62	0.00	

11/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.56	0.00	0.00	161.37	2,900,510.00 *gal*
Application event totals		0.56	0.00	0.00	161.37	

04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.71	0.00	0.00	203.07	3,650,080.00 *gal*
Application event totals		0.71	0.00	0.00	203.07	

Turkey 3/4 - 05/20/2023: Corn, silage

Field name: Turkey 3/4

Crop: Corn, silage

Plant date: 05/20/2023

Turkey 3/4 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	215.59	102.63	327.87	0.00	315.00 *ton*
Application event totals		215.59	102.63	327.87	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.27	0.00	0.00	649.09	11,667,220.00 *gal*
Application event totals		2.27	0.00	0.00	649.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.48	0.00	0.00	424.26	7,626,060.00 *gal*
Application event totals		1.48	0.00	0.00	424.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.30	0.00	0.00	371.68	6,680,950.00 *gal*
Application event totals		1.30	0.00	0.00	371.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.07	0.00	0.00	304.60	5,475,120.00 *gal*
Application event totals		1.07	0.00	0.00	304.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	308.23	5,540,300.00 *gal*
Application event totals		1.08	0.00	0.00	308.23	

Turkey 3/4 - 05/20/2023: Corn, silage

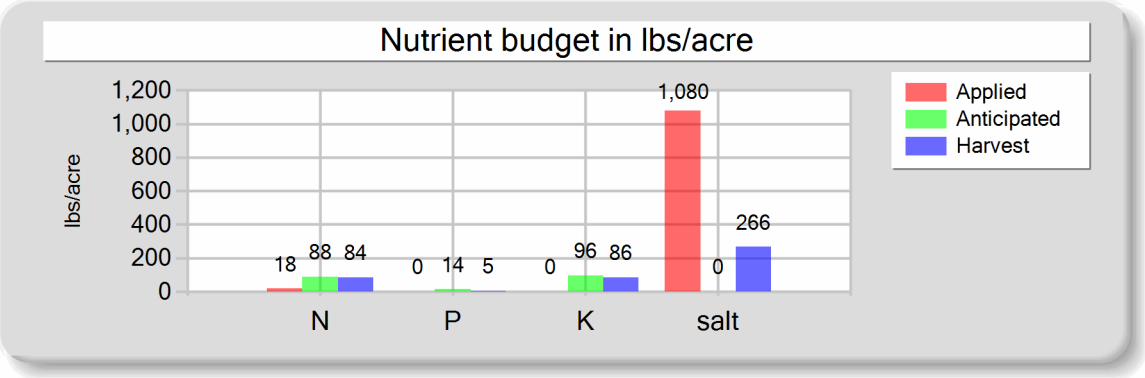
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	348.11	6,257,280.00 *gal*
Application event totals		1.22	0.00	0.00	348.11	

B. NUTRIENT BUDGET

Alfreds - 11/01/2015: Almond, in shell

Field name: Alfreds Crop: Almond, in shell Plant date: 11/01/2015



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	3.78	0.00	0.00	1,079.70
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	17.78	0.00	0.00	1,079.70
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	83.98	5.32	85.88	266.00
Nutrient balance	-66.20	-5.32	-85.88	813.70
Applied to removed ratio	0.21	0.00	0.00	4.06

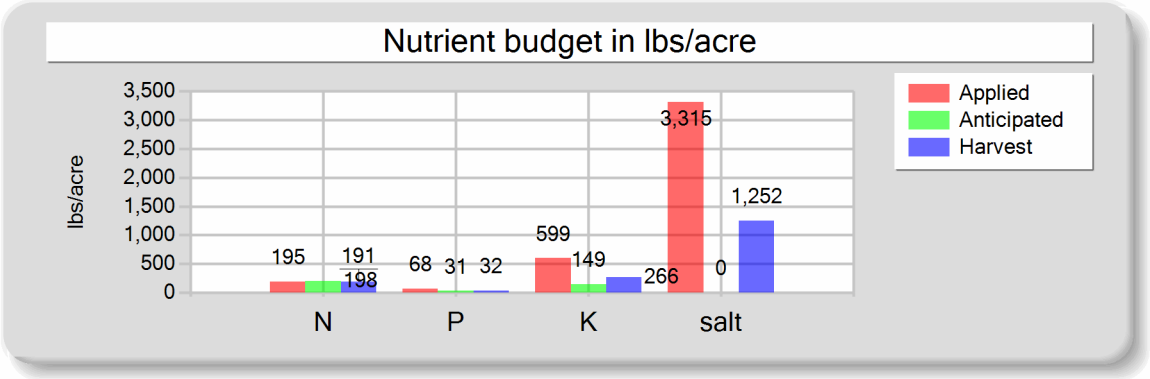
Fresh water applied	
25,876,460.00	*gallons*
952.94	*acre-inches*
23.82	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home - 11/01/2022: Wheat, silage, soft dough

Field name: Home | Crop: Wheat, silage, soft dough | Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.36	67.75	599.09	2,794.53
Fresh water	1.82	0.00	0.00	520.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.18	67.75	599.09	3,314.89
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	190.64	32.25	265.57	1,251.98
Nutrient balance	4.54	35.50	333.52	2,062.91
Applied to removed ratio	1.02	2.10	2.26	2.65

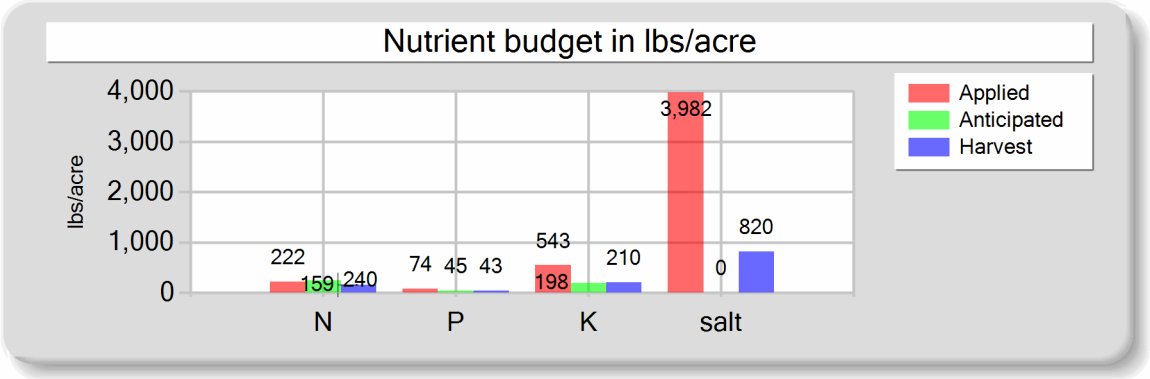
Fresh water applied	
18,706,660.00	*gallons*
688.90	*acre-inches*
11.48	*inches/acre*

Process wastewater applied	
5,400,000.00	*gallons*
198.86	*acre-inches*
3.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home - 05/20/2023: Corn, silage

Field name: Home | Crop: Corn, silage | Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.79	74.20	542.93	1,668.58
Fresh water	8.10	0.00	0.00	2,313.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.88	74.20	542.93	3,982.09
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	158.86	42.70	210.10	819.90
Nutrient balance	63.03	31.50	332.83	3,162.18
Applied to removed ratio	1.40	1.74	2.58	4.86

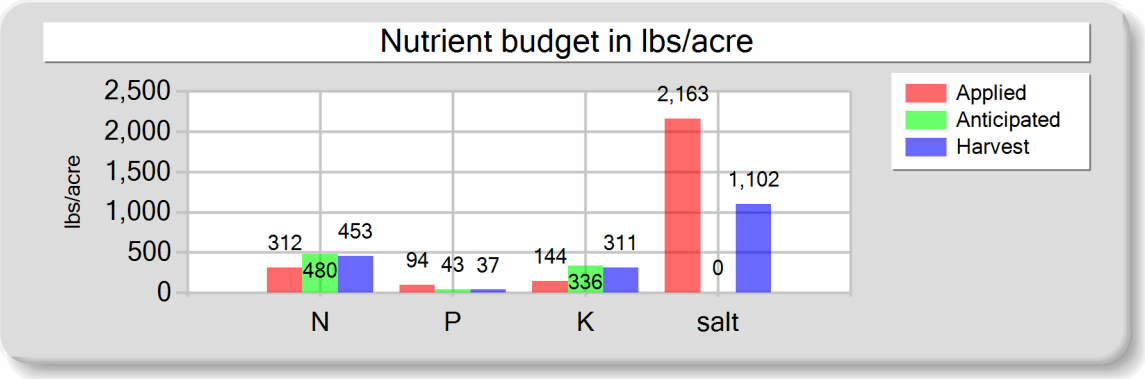
Fresh water applied	
83,169,680.00	*gallons*
3,062.86	*acre-inches*
51.05	*inches/acre*

Process wastewater applied	
3,600,000.00	*gallons*
132.58	*acre-inches*
2.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

Salsa Ranch - 01/15/2023: Alfalfa, hay

Field name: Salsa Ranch Crop: Alfalfa, hay Plant date: 01/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	290.20	94.08	144.44	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.57	0.00	0.00	2,163.16
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	311.77	94.08	144.44	2,163.16
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	452.65	36.74	310.95	1,102.10
Nutrient balance	-140.87	57.35	-166.51	1,061.06
Applied to removed ratio	0.69	2.56	0.46	1.96

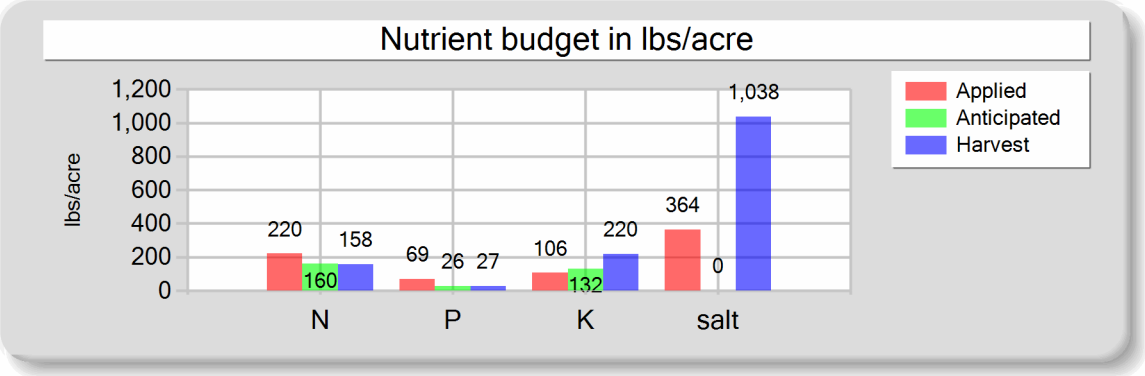
Fresh water applied	
101,094,180.00	*gallons*
3,722.95	*acre-inches*
47.73	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Turkey 3/4 - 11/01/2022: Wheat Hay

Field name: Turkey 3/4 Crop: Wheat Hay Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	212.21	68.80	105.62	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.28	0.00	0.00	364.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.49	68.80	105.62	364.43
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	157.99	26.72	220.08	1,037.52
Nutrient balance	62.50	42.08	-114.46	-673.09
Applied to removed ratio	1.40	2.57	0.48	0.35

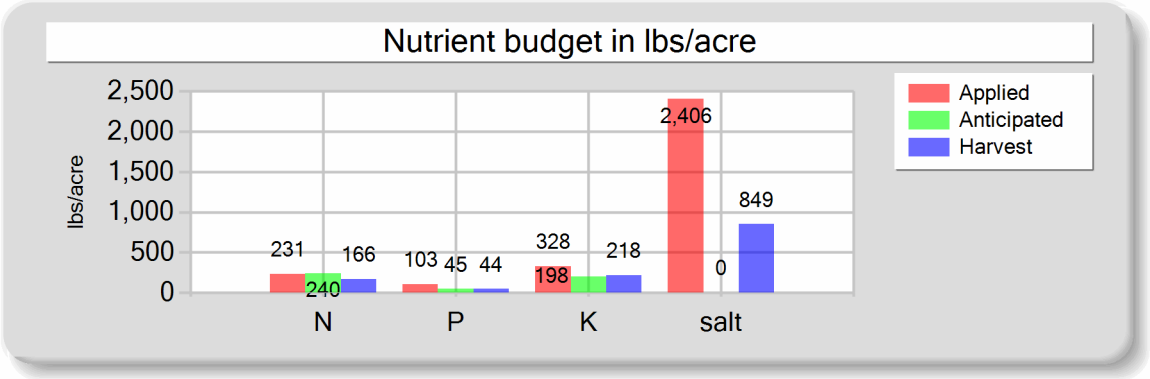
Fresh water applied	
6,550,590.00	*gallons*
241.24	*acre-inches*
8.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Turkey 3/4 - 05/20/2023: Corn, silage

Field name: Turkey 3/4 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	215.59	102.63	327.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.42	0.00	0.00	2,405.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.02	102.63	327.87	2,405.97
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	166.33	44.24	217.64	849.34
Nutrient balance	64.69	58.39	110.22	1,556.63
Applied to removed ratio	1.39	2.32	1.51	2.83

Fresh water applied	
43,246,930.00	*gallons*
1,592.64	*acre-inches*
53.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 SSB Manure

Sample and source description: 2022 SSB Manure

Sample date: 10/05/2022 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,900.00	7,100.00	10,900.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,300.00	14,900.00	47,600.00							
DL	0.01	0.01	0.01							

Corral Manure - Fall

Sample and source description: Corral Manure - Fall

Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 19.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,500.00	11,100.00	38,400.00	24,700.00	13,700.00	6,500.00	6,700.00	10,000.00		47.50
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

B. PROCESS WASTEWATER ANALYSES

2022 Lagoon Q4

Sample and source description: 2022 Lagoon Q4

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	298.00	222.00	0.00	1.70	70.90	723.00								6,970.00	3,400
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	267.00	146.00	0.00	1.50	78.80	856.00								6,220.00	3,670
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	210.00	129.00	0.00	1.30	104.00	851.00								6,260.00	3,950
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	521.00	176.00	0.00	0.60	172.00	1,210.00								8,040.00	3,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	288.00	225.00	0.00	0.00	48.40	864.00								8,050.00	4,030
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

C. FRESH WATER ANALYSES

Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.70								328.00	200
DL		0.01	0.01								0.01	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.70	0.00	11.70								1,370.00	
DL	0.01	0.01	0.01								0.01	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Home - 05/20/2023: Corn, silage

Lima Corn

Sample and source description: Lima Corn

Sample date: 10/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	2,500.00	12,300.00		4.80
DL	0.01	0.01	0.01		1.00

Salsa Ranch - 01/15/2023: Alfalfa, hay

Salsa Alfalfa

Sample and source description: Salsa Alfalfa

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,500.00	2,800.00	23,700.00		8.40
DL	0.01	0.01	0.01		1.00

Turkey 3/4 - 05/20/2023: Corn, silage

Lima Corn

Sample and source description: Lima Corn

Sample date: 10/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	2,500.00	12,300.00		4.80
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

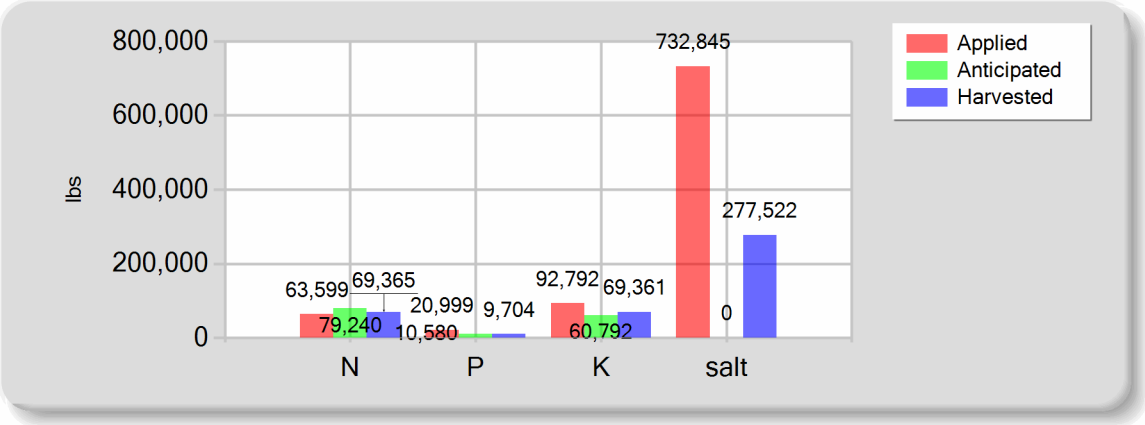
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

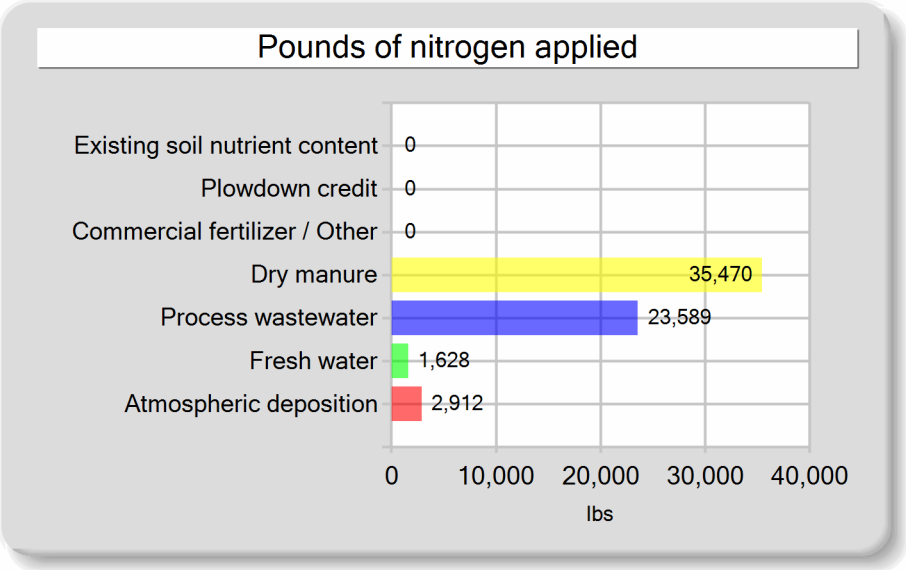
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

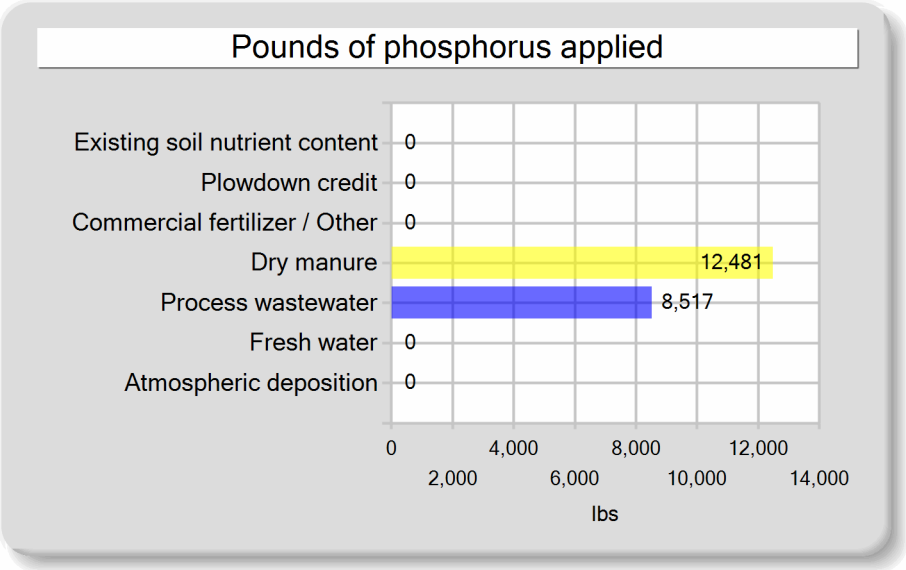
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	35,470.00	12,481.47	24,270.93	0.00
Process wastewater	23,588.83	8,517.28	68,521.30	267,786.88
Fresh water	1,627.70	0.00	0.00	465,057.67
Atmospheric deposition	2,912.00	0.00	0.00	0.00
Total nutrients applied	63,598.53	20,998.75	92,792.23	732,844.55
Anticipated crop nutrient removal	79,240.00	10,579.60	60,792.00	0.00
Actual crop nutrient removal	69,365.11	9,704.15	69,361.25	277,522.47
Nutrient balance	-5,766.58	11,294.60	23,430.98	455,322.08
Applied to removed ratio	0.92	2.16	1.34	2.64

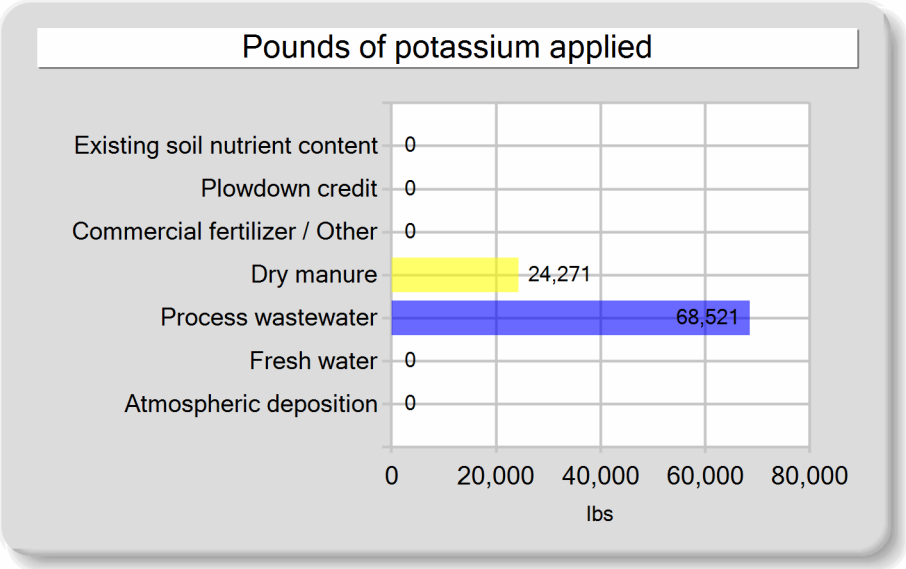
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

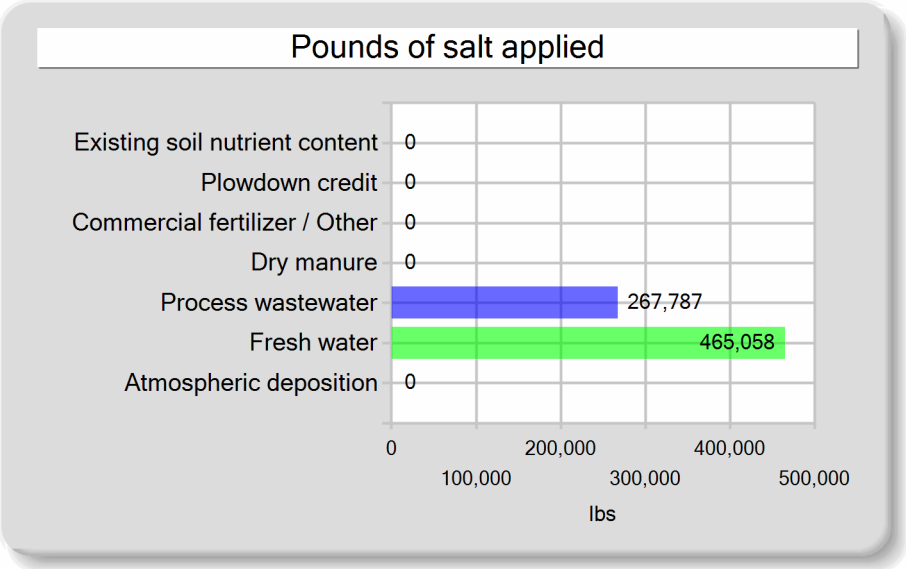


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Facility is not housing milk cows, just support stock so minimal lagoon water and manure is being produced

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Alfred or Lucy Nunes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 1, 2024	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Double L & N Dairy
PO Box 494
Gustine CA 95322
00-0021365 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1015
Sampled By: M Dekeyrel
Requested By: Lucy Nunes
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/13/23	12:23	1370	1.37	11.7	ND		7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Double L & N Dairy
PO Box 494
Gustine CA 95322
00-0021365 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1015
Sampled By: M Dekeyrel
Requested By: Lucy Nunes
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/13/23	12:23	**1370**	**1.37**	**11.7**	31.94	2.66

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Double L & N Dairy
PO Box 494
Gustine, CA 95322

Account# 00-0021365
Account Manager: Ben Nydam
Submitted By: Lucy Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:44

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L1015-01	DW - MB	Ag Water	M Dekeyrel		12/13/2023 12:23

Default Cooler Temperature on Receipt °C: 4.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Double L & N Dairy
PO Box 494
Gustine, CA 95322

Account# 00-0021365
Account Manager: Ben Nydam
Submitted By: Lucy Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:44

Sample Results

Sample: DW - MB
23L1015-01 (Water)

Sampled: 12/13/2023 12:23
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.37**	mmhos/cm	0.01	1		12/19/23 11:59	SM 2510 B		BEL0706
Electrical Conductivity umhos	**1370**	umhos/cm	10.0	1		12/19/23 11:59	SM 2510 B		BEL0706
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:23	Field		BEL0655
Nitrate Nitrogen as NO3N	**11.7**	mg/L	0.1	1	10	12/19/23 00:01	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:59	SM 4500-H+	H	BEL0706
pH	**7.6**	units	1.0	1		12/19/23 11:59	SM 4500-H+	H	BEL0706

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DELLAVALLE™
LABORATORY INC

Double L & N Dairy
PO Box 494
Gustine, CA 95322

Account# 00-0021365
Account Manager: Ben Nydam
Submitted By: Lucy Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:44

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0686										
Blank (BEL0686-BLK1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0686-BS1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
LCS (BEL0686-BS2)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEL0686-DUP1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L		5.1			0.974	10
Duplicate (BEL0686-DUP2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L		5.3			0.530	10
Matrix Spike (BEL0686-MS1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.1	106	90-110		
Matrix Spike (BEL0686-MS2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.3	102	90-110		
Reference (BEL0686-SRM1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0686-SRM2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0686-SRM3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™ LABORATORY INC

Double L & N Dairy
PO Box 494
Gustine, CA 95322

Account# 00-0021365
Account Manager: Ben Nydam
Submitted By: Lucy Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:44

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706										
Blank (BEL0706-BLK1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0706-BLK2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.9		1.0	units						
Blank (BEL0706-BLK3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.8		1.0	units						
Duplicate (BEL0706-DUP1)		**Source: 23L1017-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1.74		0.01	mmhos/cm		1.73			0.403	10
pH	7.7		1.0	units		7.8			0.258	10
Electrical Conductivity umhos	1740		10.0	umhos/cm		1730			0.403	10
Duplicate (BEL0706-DUP2)		**Source: 23L1088-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	0.52		0.01	mmhos/cm		0.52			0.116	10
Electrical Conductivity umhos	517		10.0	umhos/cm		517			0.116	10
pH	6.7		1.0	units		6.8			0.297	10
Reference (BEL0706-SRM1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	440			umhos/cm	426.0		103	90-110		
Reference (BEL0706-SRM2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0706-SRM3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEL0706-SRM4)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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DELLAVALLE™
LABORATORY INC

Double L & N Dairy
PO Box 494
Gustine, CA 95322

Account# 00-0021365
Account Manager: Ben Nydam
Submitted By: Lucy Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:44

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706 (Continued)										
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM6)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM7)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM8)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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12/14/23 14:20 23L1015

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21365 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Double L&N Dairy
Address: 28471 W Hulen Road
City, State, Zip: Gustine, CA 95322
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Lucy Nunes - (209) 564-1054

Ranch _____

Date sampled: 12/13/23

Sampled by: M. Dekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 1 No. Bottles: 1

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test**)
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-MB	Sampled From:	12/13	12:23pm	Ø	4.3
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M. Dekeyrel	CAS Inc	12/13/23 @ 12:23pm	12/14/23 12p
Second	LSM	DLI	12/14/23 12pm	12/14/23 2:15pm
Third				
Fourth	[illegible]	DLI	12/14 1420	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L1015

SEND OUT